

Modeling viral kinetics

2

5

**Infectious
exposure**



Infinite possibilities

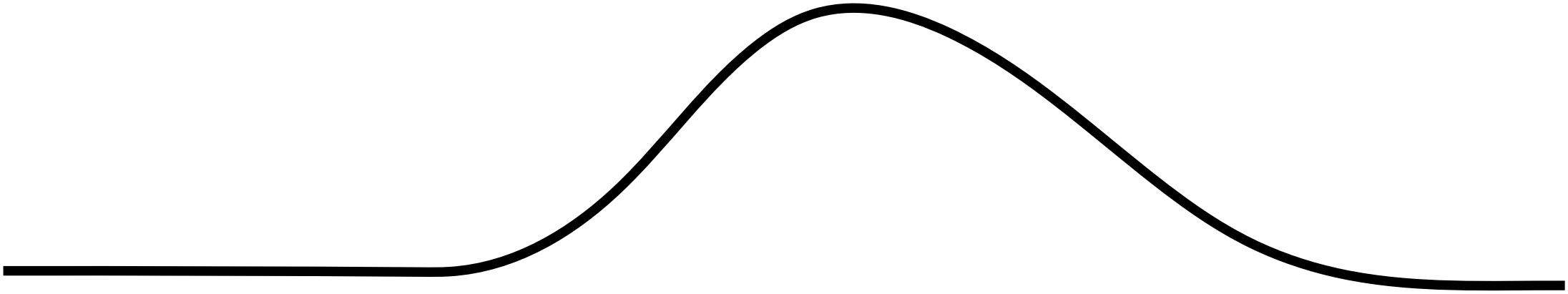
symptoms











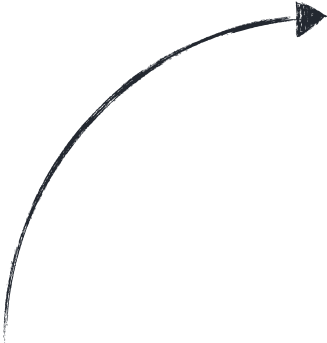




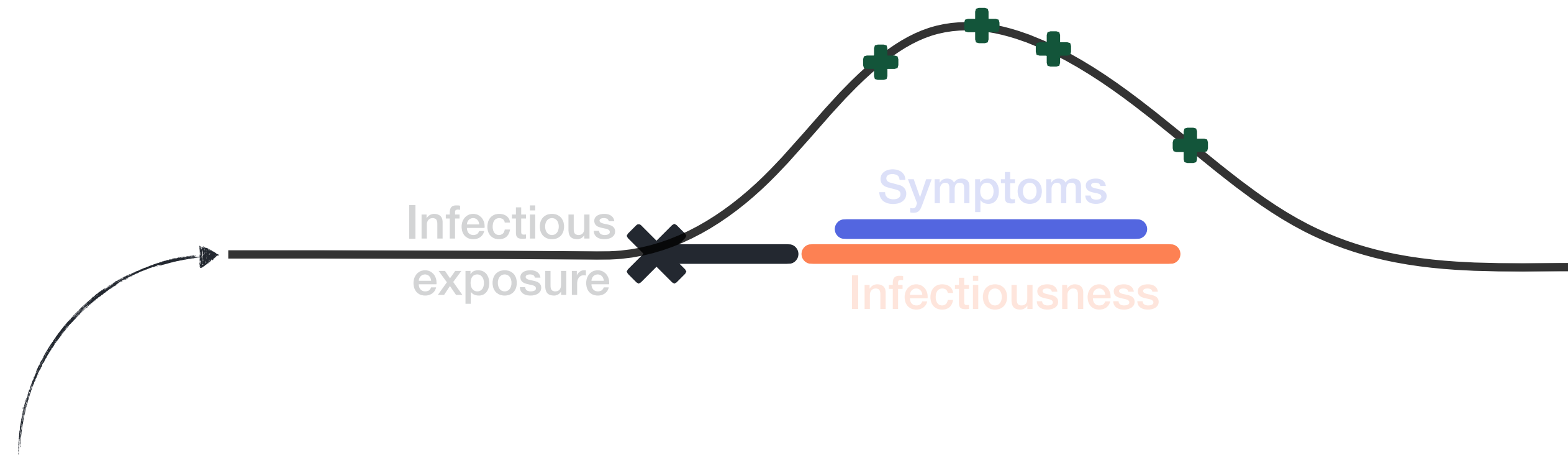




Models are what all must make use of quantitative (and qualitative) testing data



Modeling viral kinetics



Models are what allow us to make use of quantitative (and qualitative!) testing data

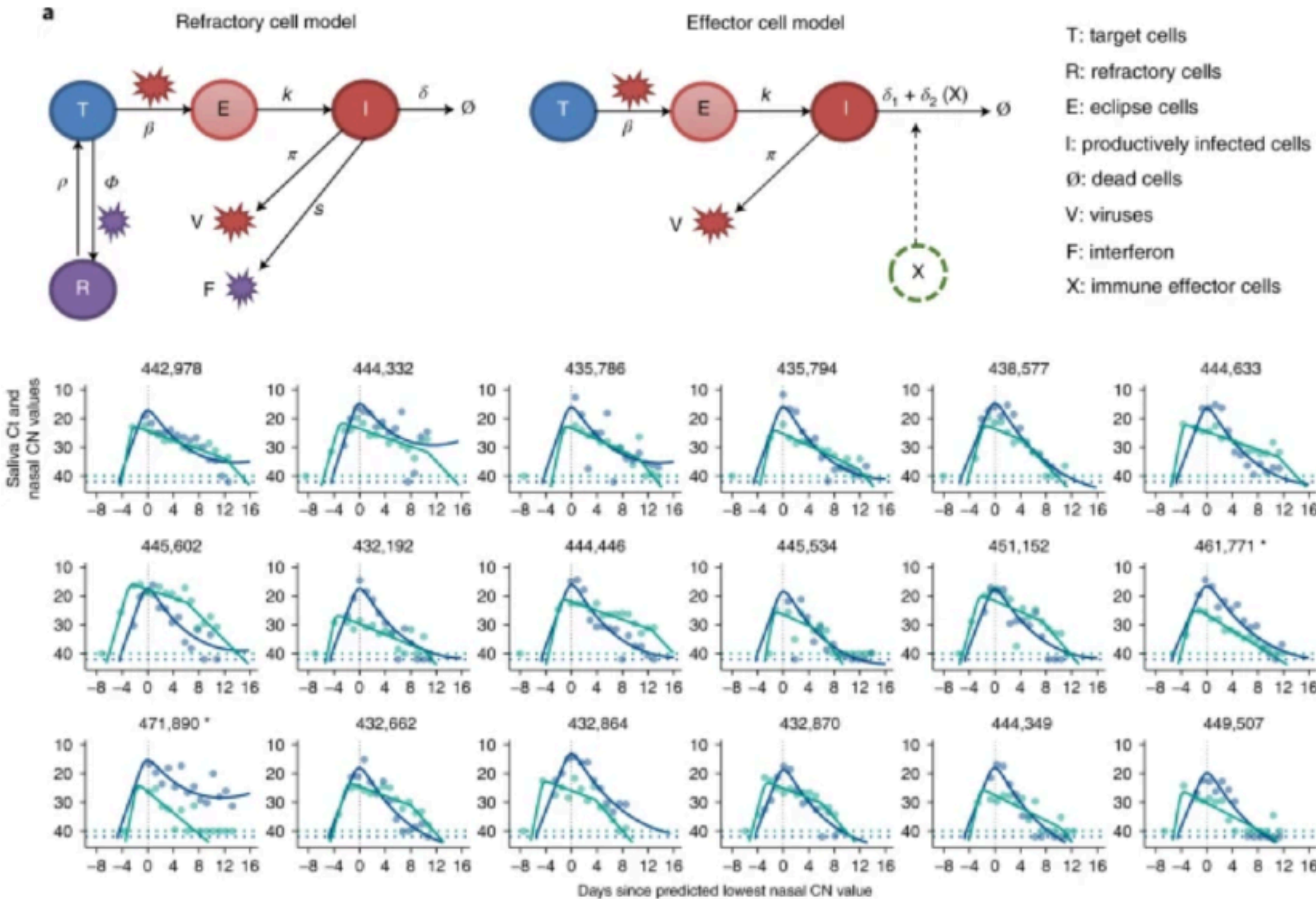
Modeling viral kinetics

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Daily longitudinal sampling of SARS-CoV-2 infection reveals substantial heterogeneity in infectiousness

Ruian Ke, Pamela P. Martinez, Rebecca L. Smith, Laura L. Gibson, Agha Mirza, Madison Conte, Nicholas Gallagher, Chun Huai Luo, Junko Jarrett, Ruifeng Zhou, Abigail Conte, Tongyu Liu, Mireille Farjo, Kimberly K. O. Walden, Gloria Rendon, Christopher J. Fields, Leyi Wang, Richard Fredrickson, Darci C. Edmonson, Melinda E. Baughman, Karen K. Chiu, Hannah Choi, Kevin R. Scardina, Shannon Bradley, Stacy L. Gloss, Crystal Reinhart, Jagadeesh Yedetore, Jessica Quicksall, Alyssa N. Owens, John Broach, Bruce Barton, Peter Lazar, William J. Heetderks, Matthew L. Robinson, Heba H. Mostafa, Yukari C. Manabe, Andrew Pekosz, David D. McManus & Christopher B. Brooke

Nature Microbiology 7, 640–652 (2022) | [Cite this article](#)



Viral kinetics of sequential SARS-CoV-2 infections

Stephen M. Kissler, James A. Hay, Joseph R. Fauver, Christina Mack, Caroline G. Tai, Deverick J. Anderson, David D. Ho, Nathan D. Grubaugh & Yonatan H. Grad

Nature Communications 14, Article number: 6206 (2023) | [Cite this article](#)

$$\omega_{p[i]} = \text{Exp} \left[\beta_p + \beta_{p[c]} + \sum_a \beta_{p[a]} + \tau_p \eta_{p[i]} \right] \omega_p^*$$

$$\omega_{r[i]} = \text{Exp} \left[\beta_r + \beta_{r[c]} + \sum_a \beta_{r[a]} + \tau_r \eta_{r[i]} \right] \omega_r^*$$

$$\delta_{[i]} = \text{Exp} \left[\beta_\delta + \beta_{\delta[c]} + \sum_a \beta_{\delta[a]} + \tau_\delta \eta_{\delta[i]} \right] \delta^*$$

